

1. Launch one EC2 using Amazon Linux 2 image and add a script in user data to install Apache.

The screenshot shows the AWS Management Console 'Launch Instance' wizard. The 'Summary' tab is active, displaying instance details for 'Amazon Linux 2023'. The 'Application and OS Images (Amazon Machine Image)' section is expanded, showing 'Amazon Linux 2023 AMI' as the selected image. The 'Summary' section on the right shows '1 instance' and a 'Launch Instance' button.

[EC2](#) > [Instances](#) > Launch an Instance
Launch EC2 instance

Metadata response hop limit: info

Allow tags in metadata: info

Select

User data - optional: info

Upload a file with your user data or enter it in the field.

+ Choose file


```
#bin/bash
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd

echo "Hi!Apache Web Server - Task 1"/>

```

Summary

Number of instances: info

1

Software Image [AMI]

Amazon Linux 2023 Amazon Linux 2...read more
ami-032e56c767da0db

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volume)

1 volume(s) - 8 GB

Cancel

Launch instance

Preview code

☐ User data has already been base64 encoded

Instances

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
Apache	i-0528ea7f122c0ffa	Running	m3.xlarge	Initialising	us-east-1d	ec2-54-210-42-36.us-east-1.amazonaws.com

I-0528ea7f122c0ffa (Apache)

- Details
- Status and alarms
- Monitoring
- Security
- Networking
- Storage
- Tags

▼ Instance summary

Instance ID	Public IPv4 address	Private IPv4 addresses
i-0528ea7f122c0ffa	54.210.42.36 open address ↗	172.31.100.91

IPv4 address

Instance state	Public DNS
Running	ec2-54-210-42-36.compute-1.amazonaws.com open address ↗

Hostname type

Private IP DNS name (IPv4 only)

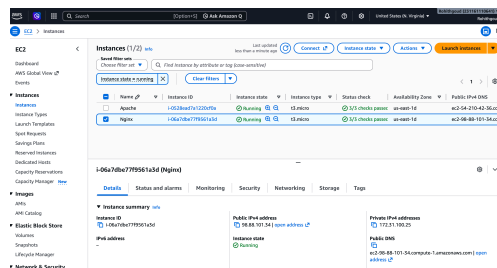
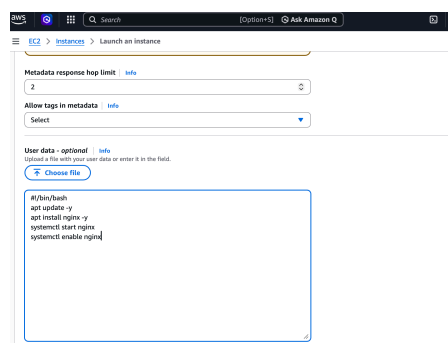
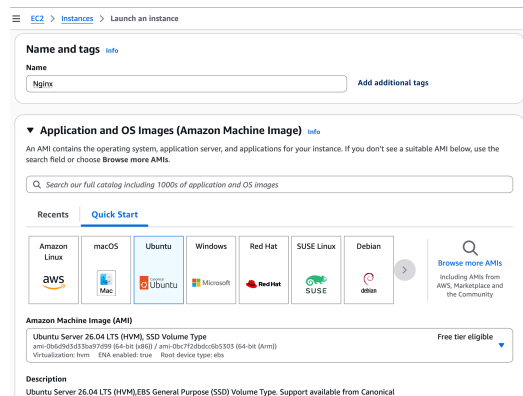
```

[ec2-user@ip-172-31-100-91 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-100-91 ~]$ sudo systemctl enable httpd
[ec2-user@ip-172-31-100-91 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Mon 2024-08-31 11:21:42 UTC; 16min ago
     Docs: man:httpd.service(8)
  Main PID: 3521 (httpd)
    Status: "Total requests: 0; idle/busy workers 100/0;Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 177 (limit: 1059)
   Memory: 13.6M
      CPU: 913ms
   CGroup: /system.slice/httpd.service
           └─3521 /usr/sbin/httpd -DFOREGROUND
           └─3623 /usr/sbin/httpd -DFOREGROUND
           └─3629 /usr/sbin/httpd -DFOREGROUND
           └─3630 /usr/sbin/httpd -DFOREGROUND
           └─3694 /usr/sbin/httpd -DFOREGROUND

Aug 31 11:21:42 ip-172-31-100-91.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Aug 31 11:21:42 ip-172-31-100-91.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
[ec2-user@ip-172-31-100-91 ~]$
```

Apache Web Server - Task 1

2. Launch one EC2 using Ubuntu image and add a script in user data to install Nginx.



```
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Running: /usr/lib/systemd/systemd-sysv-install enable nginx
ubuntu@ip-172-31-100-25:~$ sudo systemctl status nginx
nginx.service - A high performance web server and a reverse proxy server
Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
Active: active (running) since Mon 2024-08-31 11:57:11 UTC; 2min 5s ago
Invocation: 6277e0e08b7f0ca197f0e0a1c0c0f8
Process: Main PID: 2143 (nginx)
Tasks: 3 (limit: 4277)
Memory: 3M (peak: 7.2M)
CPU: 44ms
CGroup: /system.slice/nginx.service
└─2143 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
   └─2144 "nginx: worker process"
      └─2147 "nginx: worker process"
```

Aug 31 11:57:11 ip-172-31-100-25 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...

Aug 31 11:57:11 ip-172-31-100-25 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.

ubuntu@ip-172-31-100-25:~\$

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

3. Launch one Windows server and install Tomcat on Windows.

The screenshot shows the AWS Management Console 'Instances' page. The top navigation bar includes the AWS logo, search bar, and account information. The left sidebar shows the 'EC2' menu. The main content area displays a table of instances. One instance is listed: 'Windows-Tomcat' with ID 'i-038e973d07022aa32', in a 'Running' state, using the 't3.micro' instance type. The instance is in the 'us-east-1d' Availability Zone. The console shows the 'Launch instances' button and a table of instance details.

The screenshot shows the 'Connect to Windows instance' page in the AWS console. The page provides instructions on how to connect to the instance using RDP (Remote Desktop Protocol). The 'How to connect' section lists a 6-step process: 1. Open or install a remote desktop client, 2. Download the RDP shortcut file, 3. Open the RDP client and run the shortcut file, 4. Retrieve password, 5. Enter username and password, 6. Connect. It also shows the 'RDP shortcut file' download link and the 'Public DNS' address: ec2-98-88-101-34.compute-1.amazonaws.com. The 'Username' is 'Administrator' and the 'Password' is 'Get Windows password'.

The screenshot shows the 'Get Windows password' page in the AWS console. It displays the 'Private key' field with the contents of the 'windows.pem' file. The page includes a 'Decrypt password' button.

The screenshot shows the Apache Tomcat 10.1.59 web interface. The page displays the 'Home' tab with a 'Congratulations!' message for successful installation. It includes links to 'Recommended Reading' (Security Considerations, Manager Application, Cluster/Session Replication), 'Developer Quick Start' (Tomcat Setup, First Web Application, Resins & AAA, JDBC Data Sources, Examples, Servlet Specifications, Tomcat Versions), and 'Managing Tomcat' (For security, access to the manager webapp is restricted, Users are defined in \$CATALINA_HOME/conf/tomcat-users.xml, In Tomcat 10.1 access to the manager application is split between different users, Read more...). It also includes a 'Getting Help' section with links to 'FAQ and Mailing Lists' and 'Tomcat announcement'.

4. Take a snapshot of the instance created in Task 1.

Instances (1/3)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
Windows-Tom...	i-038e973d07022aa52	Running	t3.micro	3/3 checks passed	us-east-1d	ec2-98-88-101-34.cor
Apache	i-0528ead7a1220cf0a	Running	t3.micro	3/3 checks passed	us-east-1d	ec2-54-210-42-36.cor
Nginx	i-06a7dbe77f9561a5d	Running	t3.micro	3/3 checks passed	us-east-1d	-

I-0528ead7a1220cf0a (Apache)

Volume ID	Device name	Volume size (GiB)	Volume State	Attachment status	EBS card index	Attachmen
vol-0b89d4dd1c60b9e9e	/dev/xvda	8	In-use	Attached	0	2026/08/3

Volume monitoring (1)

Alarm recommendations: 1

Average read latency (...): 1

Average write latency (...): 1.79

Read throughput (KiB/s): 1

Create snapshot

Create a point-in-time snapshot to back up the data on an Amazon EBS volume to Amazon S3.

Source volume

Volume ID: vol-0b89d4dd1c60b9e9e

Availability Zone: us-east-1d

Snapshot details

Description: Add a description for your snapshot

255 characters maximum.

Encryption: Not encrypted

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

You can add up to 50 tags.

Cancel Create snapshot

Volumes (1)

Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume ID
	vol-0b89d4dd1c60b9e9e	gp3	8 GiB	3000	125	snap-0bc9c3f3f...	-

Fault tolerance for all volumes in this Region

Snapshot summary

Last updated on Tue, Sep 01, 2026, 12:25:43 AM (GMT+05:30)

Snapshots to back up volumes can be manually created, or created through automation such as Amazon Data Lifecycle Manager, AWS Backup, custom scripts, or third-party tools.

Volumes backed up in the last 24 hours / Total # volumes: 0 / 3

Data Lifecycle Manager default policy for EBS Snapshots status: No default policy set up | Create policy

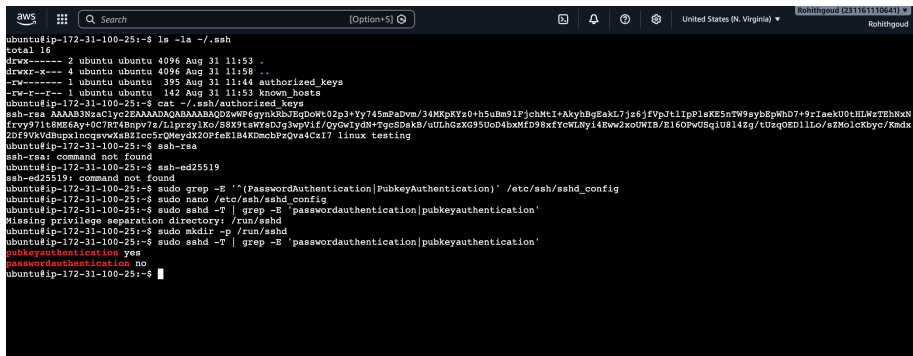
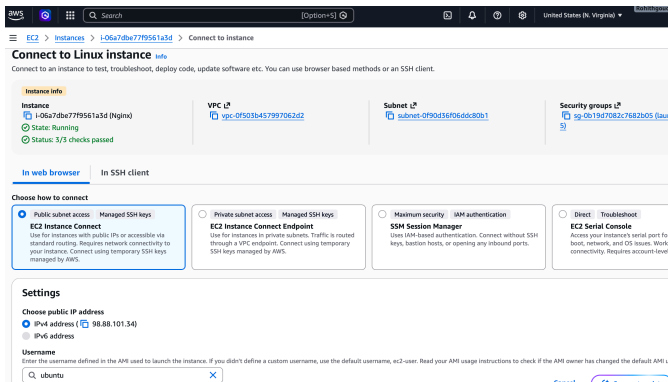
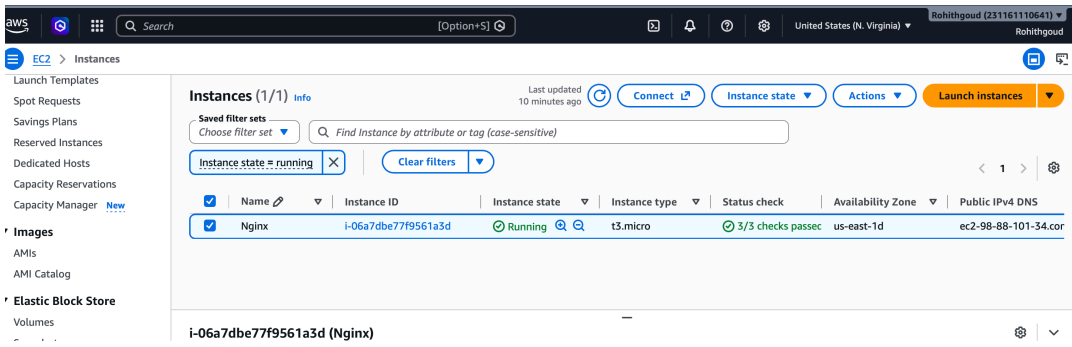
Snapshots (1/1)

Name	Snapshot ID	Full snapshot size	Volume size	Description	Storage tier	Snapshot size
	snap-0e9021f9e1112333a	1.76 GiB	8 GiB	Task-4-Snapshot-Amazon...	Standard	Complete

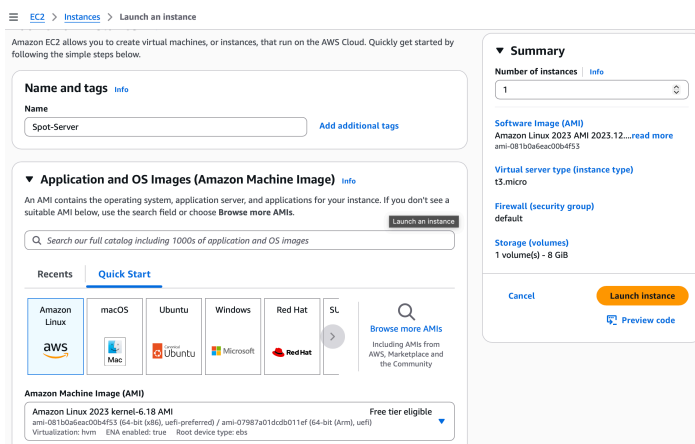
Snapshot ID: snap-0e9021f9e1112333a

Details	Snapshot settings	Storage tier	Tags
Snapshot ID: snap-0e9021f9e1112333a	Full snapshot size: 1.76 GiB	Progress: 100%	Snapshot status: Completed
Owner: 231161110641	Started: Tue Sep 01 2026 09:26:58 GMT+05:30 (India Standard Time)	Product codes: -	Fast snapshot restore: -
Description: Task-4-Snapshot-AmazonLinux-Apache			

5. Assign passwordless authentication for the EC2 created in Task 2.



6. Launch any EC2 using the spot purchasing option.



EC2 > Instances > Launch an instance

Placement group: [Info](#)
 Select [Create new placement group](#)

EBS-optimized instance: [Info](#)
 Enable

Instance bandwidth configuration: [Info](#)
 Select

Purchasing option: [Info](#)
☐ None
☐ Capacity Blocks
 Launch instances for your active capacity blocks
☐ Interruptible Capacity Reservations
 Launch instances for your interruptible Capacity Reservations
☒ Spot instances
 Request Spot instances at the Spot price, capped at the On-Demand price
[Customize Spot instance options](#)

Capacity reservation: [Info](#)
 Select

Tenancy: [Info](#)
 Select

RAM disk ID: [Info](#)
 Select

Summary

Number of instances: [Info](#)
 1

Software Image (AMI)
 Amazon Linux 2023 AMI 2023.12...[read more](#)
 ami-081b0a6eac00b4f53

Virtual server type (instance type)
 t3.micro

Firewall (security group)
 default

Storage (volumes)
 1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Instances (1/4) [Info](#) Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Save filter sets [Choose filter set](#) [Find Instance by attribute or tag \(case-sensitive\)](#)

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☐	Windows-Tom...	i-038e973d07022aa32	Stopped	t3.micro	-	us-east-1d	-
☒	Spot-Server	i-09634591989eea9c8	Running	t3.micro	Initializing	us-east-1d	ec2-67-202-42-237.cc
☐	Apache	i-0528ead7a1220cfaa	Stopped	t3.micro	-	us-east-1d	-
☐	Nginx	i-06a7dbe77f9561a3d	Stopped	t3.micro	-	us-east-1d	ec2-98-88-101-34.cor

i-09634591989eea9c8 (Spot-Server)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

Instance summary [Info](#)

Instance ID i-09634591989eea9c8	Public IPv4 address 67.202.42.237 open address	Private IPv4 addresses 172.31.100.117
IPv6 address	Instance state Running	Public DNS

7. Enable termination policy on the EC2 created in Task 2.

Instances (1/1) [Info](#) Last updated 1 minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Save filter sets [Choose filter set](#) [Find Instance by attribute or tag \(case-sensitive\)](#)

Instance state: running [Clear filters](#)

☒	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☒	Nginx	i-06a7dbe77f9561a3d	Running	t3.micro	3/3 checks passed	us-east-1d	ec2-98-88-101-34.cor

i-06a7dbe77f9561a3d (Nginx)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

Instance summary [Info](#)

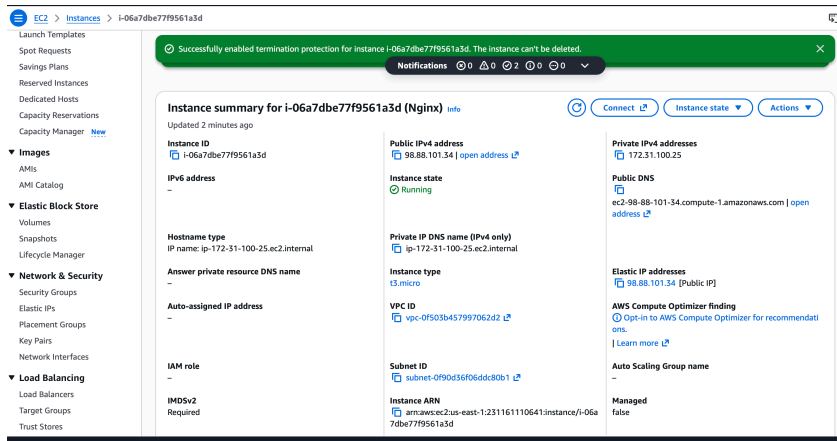
Instance ID i-06a7dbe77f9561a3d	Public IPv4 address 98.88.101.34 open address	Private IPv4 addresses 172.31.100.25
IPv6 address	Instance state Running	Public DNS ec2-98-88-101-34.compute-1.amazonaws.com open address
Hostname type IP name: ip-172-31-100-25.ec2.internal	Private IP DNS name (IPv4 only) ip-172-31-100-25.ec2.internal	Elastic IP addresses 98.88.101.34 [Public IP]

Successfully removed termination protection for instance i-06a7dbe77f9561a3d. The instance can be deleted.

Instance summary for i-06a7dbe77f9561a3d (Nginx) [Info](#) [Connect](#) [Instance state](#) [Actions](#)

Updated 1 minute ago

Instance ID i-06a7dbe77f9561a3d	Public IPv4 address 98.88.101.34 open address	Private IPv4 addresses 172.31.100.25
IPv6 address	Instance state Running	Public DNS ec2-98-88-101-34.compute-1.amazonaws.com open address
Hostname type IP name: ip-172-31-100-25.ec2.internal	Private IP DNS name (IPv4 only) ip-172-31-100-25.ec2.internal	Elastic IP addresses 98.88.101.34 [Public IP]
Answer private resource DNS name	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommended ops. Learn more
Auto-assigned IP address	VPC ID vpc-9f503b457997062d3 open	Auto Scaling Group name
IAM role	Subnet ID subnet-9f9d56f6966de80b1 open	Managed
Instance ARN arn:aws:ec2:us-east-1:231161110641:instance/i-06a7dbe77f9561a3d	Instance ARN arn:aws:ec2:us-east-1:231161110641:instance/i-06a7dbe77f9561a3d	Managed
Owner		



8. Launch one EC2 using AWS CLI.

```
aws
Search

ubuntu@ip-172-31-100-60:~$ whoami
ubuntu
ubuntu@ip-172-31-100-60:~$ aws --version
Command 'aws' not found, but can be installed with:
sudo apt install awscli
ubuntu@ip-172-31-100-60:~$
```

```
aws
Search [Option+S]

Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Tue Sep  1 08:05:10 UTC 2026

System load:  0.0      Temperature:   -273.1 C
Usage of /:   36.6% of 6.61GB  Processes:    115
Memory usage: 30%      Users logged in: 0
Swap usage:  0%        IPv4 address for ens5: 172.31.100.60

Expanded Security Maintenance for Applications is not enabled.

147 updates can be applied immediately.
113 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Sep  1 07:45:28 2026 from 18.206.107.27
ubuntu@ip-172-31-100-60:~$ aws ec2 describe-instances \
--filters "Name=tag:Name,Values=CLI-Server" \
--query 'Reservations[*].Instances[*].[InstanceId,State.Name,PublicIpAddress]' \
--output table
-----
DescribeInstances
-----
i-022f24c3a1f3acbb0  running  None
-----
ubuntu@ip-172-31-100-60:~$
```